

Agronomy & Agricultural Science

Undergraduate Core Notes: Soil Science, Crop Nutrition & Sustainable Agriculture

LEVEL: B.Sc. Agriculture COURSE: Soil Science & Crop Management MODULE: 1.0 - Nutrient Dynamics

MODULE 1: SOIL PHYSICS & NUTRIENT DYNAMICS

Soil represents a dynamic four-phase system (45% mineral matter, 25% water, 25% air, 5% organic matter). Nutrient availability depends on soil pH, Cation Exchange Capacity (CEC), and soil microbial activity.

Soil Property: Cation Exchange Capacity (CEC)

CEC measures the total capacity of a soil to hold exchangeable cations (e.g., Ca^{2+} , Mg^{2+} , K^+ , NH_4^+). Expressed in cmol_c/kg , CEC prevents essential nutrient leaching into groundwater.

Essential Plant Nutrients (NPK Framework)

Element	Uptake Form	Physiological Function	Deficiency Symptom
Nitrogen (N)	NO_3^- , NH_4^+	Amino acid & chlorophyll synthesis	Chlorosis in older leaves
Phosphorus (P)	H_2PO_4^- , HPO_4^{2-}	ATP energy transfer & root development	Purple/purplish foliage margin
Potassium (K)	K^+	Stomatal regulation & osmotic balance	Marginal leaf necrosis / scorching

MODULE 2: CROP PROTECTION & INTEGRATED PEST MANAGEMENT

Economic Injury Level (EIL)

The lowest pest population density that causes economic damage equal to the cost of pest control interventions.

Economic Threshold (ET)

The pest density at which control measures must be applied to prevent population growth from reaching the EIL.

MODULE 3: IRRIGATION SYSTEMS & WATER MANAGEMENT

Efficient water application strategies optimize crop evapotranspiration (ET_c) while minimizing environmental degradation and soil salinization.

Irrigation Systems Comparison

Irrigation Method	Water Application Efficiency	Best Suited Crops	Operational Considerations
Drip / Micro-Irrigation	90% – 95%	Horticulture, Orchards, Row Crops	High initial capital; requires filtration
Sprinkler Systems	70% – 80%	Cereals, Pastures, Field Crops	Evaporative losses in high wind/heat
Furrow / Surface	40% – 60%	Rice, Sugarcane, Field Crops	High water loss; risk of waterlogging

Sustainable Practices: Conservation Agriculture

Built on three foundational pillars: **Minimum Mechanical Soil Disturbance (No-Till)**, **Permanent Organic Soil Cover (Mulching/Cover Crops)**, and **Diversified Crop Rotations**.